

Statistics & Probability



Mean, median, mode, and range

- 1 Find the mean, median, and range of the data set: 12, 18, 15, 22, 9, 18.
 - 2 A data set of 5 values has a mean of 14. If four of the values are 10, 12, 16, 19, find the fifth value.
-

Weighted averages

- 3 A class of 20 students has an average test score of 78, and another class of 30 students has an average of 85. Find the combined average score of all 50 students.
-

Basic probability

- 4 A bag contains 6 red, 4 blue, and 5 green marbles. Find the probability of drawing a blue marble.
 - 5 A fair six-sided die is rolled twice. Find the probability of rolling a 6 both times.
-

Combinations and permutations

- 6 In how many ways can a committee of 3 people be chosen from a group of 8, using combinations?
 - 7 In how many ways can first, second, and third place be awarded among 6 runners, using permutations?
-

Conditional probability and two-way tables

- 8 A survey of 250 students found: 90 like math and science, 60 like only math, 50 like only science, 50 like neither. Find the probability a student likes science, given that they like math.
-

Standard deviation and data spread

- 9 Two data sets have the same mean, but Data Set A has standard deviation 2 and Data Set B has standard deviation 9. Explain what this tells you about the two data sets.
-

Reading and interpreting data displays

- 10 A bar graph shows quarterly sales: Q1: 40, Q2: 55, Q3: 65, Q4: 50 (in thousands of dollars). Find the total annual sales and the percent of annual sales that occurred in Q3.
-

Expected value

- 11 A game costs 5 dollars to play. There is a 20% chance of winning 20 dollars and an 80% chance of winning nothing. Find the expected value of playing the game, and determine whether it is a fair game.
-

Multi-part probability and statistics problem

- 12 This question has two parts. A jar contains 5 red and 3 blue chips. Two chips are drawn without replacement. (a) Find the probability that both chips are red. (b) Find the probability that the two chips are different colors.
-

The complement rule and 'at least' probability

- 13 A fair coin is flipped 3 times. Find the probability of getting at least one heads, using the complement rule.
-

Quartiles and the five-number summary

- 14 A data set has the five-number summary: minimum 5, $Q_1 = 12$, median 18, $Q_3 = 25$, maximum 40. Find the interquartile range (IQR), and state what it measures.
-

Independent events and the multiplication rule

- 15 A spinner has 4 equal sections (red, blue, green, yellow), and a separate fair die is rolled. Find the probability of spinning red AND rolling a number greater than 4.
-

A multi-part statistics problem with grouped data

- 16 This question has two parts. A teacher records test scores for two classes: Class A (25 students, mean 82) and Class B (15 students, mean 90). (a) Find the combined mean score for all 40 students. (b) If 5 more students join Class A with an average score of 70, find the new combined mean for all 45 students in the combined group.

Probability using a tree diagram approach

- 17** A box has 4 white and 6 black balls. Two balls are drawn without replacement. Find the probability both balls are white.